

Copper-Catalyzed Highly Enantioselective Synthesis of Cyclic Allylic and Homoallylic Alcohols with Dialkylzinc Reagents

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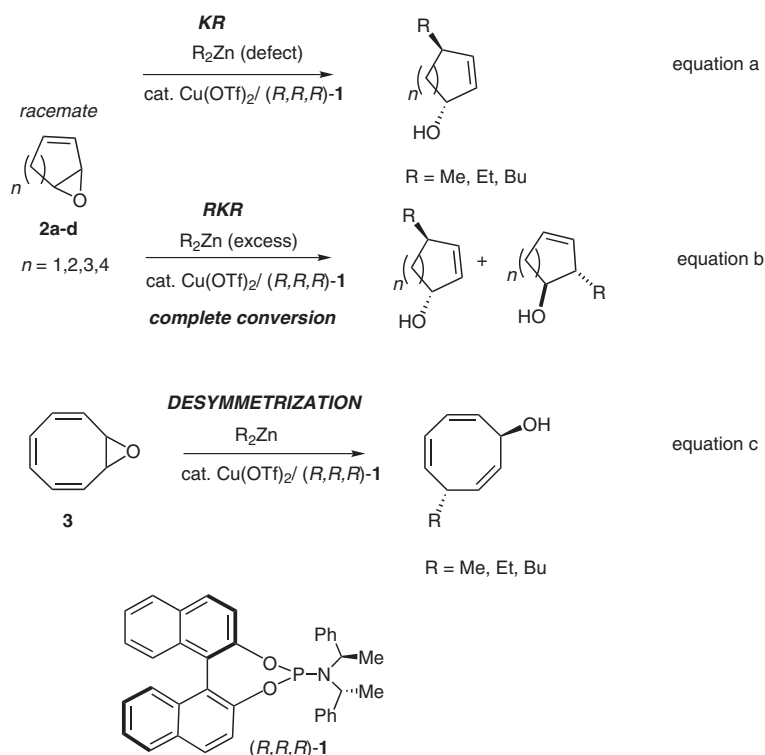
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Abstract: The copper-phosphoramidite catalyzed addition of dialkylzinc reagents to racemic or meso allylic epoxides can be synthetically exploited in different ways, depending on the substrates and reaction conditions used.

Key words: asymmetric catalysis, organozinc reagents, copper, kinetic resolution, desymmetrization



Scheme 1 Possible regio- and stereochemical outcomes of the copper-phosphoramidite catalyzed alkylation of allylic epoxides.

The development of simple, efficient procedures for the stereoselective formation of carbon-carbon bonds by the use of organometallic reagents is one of the major challenges in organic chemistry. Recently, we have introduced some new copper-catalyzed enantioselective allylic alkylation ring-opening reactions of small ring heterocycles with hard alkyl metals.¹ The regio- and stereochemical outcome of the copper-phosphoramidite catalyzed enantioselective addition of dialkylzinc reagents to cyclic allylic epoxides **2a–d** depended on the substrate and reaction conditions used (see Scheme 1). Phosphoramidite

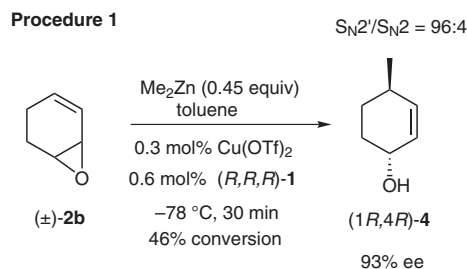
(*R,R,R*)-**1** derived from (*R*)-BINOL and (*R*)-bis-phenyl-ethylamine,² was the chosen ligand because it proved to be slightly superior to its diastereoisomer (*S,R,R*)³ with respect to the extent of regiodivergency, and the enantioselectivity of the alcohol reaction products.

When reactions were performed in accordance with a classic kinetic resolution (KR) protocol (Equation a, Scheme 1), it was possible to obtain with a high regioselectivity the corresponding allylic alcohol (*S_N2'* pathway) and to recover the unreacted allylic epoxide with a high optical purity.⁴ Very interestingly, if reactions were performed up to completion of the allylic epoxide (Equation b, Scheme 1), it was possible to obtain both the corresponding regioisomeric allylic and homoallylic alcohols (*S_N2* pathway) with a high enantioselectivity.⁵ To avoid

the inherent limitations of resolution processes, also some novel procedures for the enantioselective desymmetrization of symmetrical allylic epoxides were devised.⁶ For example, we recently demonstrated the first alkylating and enantioselective ring-opening of 1,3,5,7-cyclooctatetraene (COT) monoepoxide (**3**) to give previously unknown 4-alkylcyclooctatrienols (Equation c, Scheme 1).^{6c}

From a synthetic point of view, it should be noted that cyclic allylic- (bis-allylic-) and homoallylic alcohols with high optical purity cannot be prepared by alternative straightforward synthetic procedures. Sharpless' kinetic resolution of racemic cyclic allylic alcohols with an endocyclic double bond is known to proceed with a modest enantioselectivity.⁷ Moreover, the enantioselective epoxidation of *cis*-olefins of conjugated dienes,⁸ followed by a subsequent regioselective installation of the alkyl chain, is potentially able to give only moderate values of optical purity. On the other hand, racemic cyclic allylic epoxides **2a–d** and symmetrical COT-monoepoxide **3** can easily be prepared on a multigram scale by mono-epoxidation of the corresponding commercially available 1,3-dienes^{9a} or COT, respectively.^{9b}

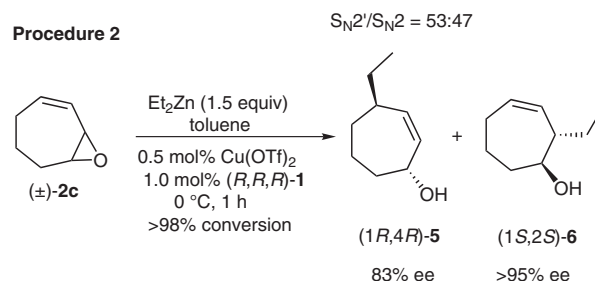
The KR protocol was recently used by our group for a novel catalytic high regio- and enantioselective synthesis of 4-methyl-2-cyclohexen-1-ol (**4**) in a multigram scale reaction (**Procedure 1**, Scheme 2).^{4b} Allylic alcohol **4** was then transformed into the corresponding α,β -unsaturated ketone by a simple oxidation reaction. Compared with other multistep syntheses of optically active 4-methyl-2-cyclohexen-1-ones, which are useful chiral building blocks,^{4b} our two-step procedure is straightforward, simple and practical. The major drawback of this protocol is the reduced degree of conversion (46%) necessary to obtain the S_N2' addition product **4** with a high regio- and enantioselectivity. However, the simple work-up procedure, based on filtration and distillation, qualifies for a further scale-up.



Scheme 2 Multigram catalytic enantioselective synthesis of 4-methyl-2-cyclohexen-1-ol via kinetic resolution of racemic 1,3-cyclohexadiene monoepoxide.^{4b}

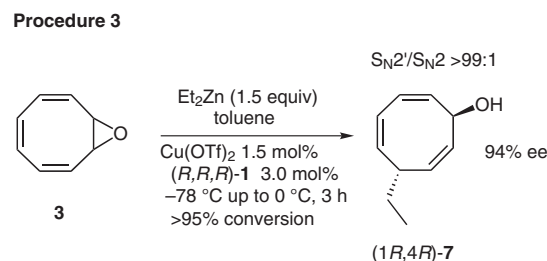
A complete conversion of a racemic allylic epoxide into constitutionally different enantiomerically enriched ring-opened products can be simply obtained by means of an excess of the dialkylzinc reagent and a longer reaction time. This is one of the very rare examples of a regiodivergent kinetic resolution (RKR) by the use of a hard organo-

metallic reagent and a chiral catalyst in a carbon-carbon bond-forming reaction.¹⁰ Very recently, we have shown that the RKR can be successfully applied to a variety of semicyclic rigid allylic epoxides and, in some specific cases, also to conformationally mobile allylic epoxides.¹¹ Like most catalytic asymmetric reactions, our RKR protocol is substrate-dependent and 1,3-cycloheptadiene monoepoxide proved to be an ideal substrate. Evidently, with this substrate, the asymmetric matching of the chiral ligand with the enantiomers of the substrate is considerable. Also by the use of reduced amounts of the chiral copper catalyst (0.5 mol%) in a gram scale reaction entirely performed at 0°C , regioisomeric alcohols **5** and **6**, having opposite configurations at the hydroxyl group-bearing carbon, were obtained in almost equal amounts, with high enantiomeric excesses (**Procedure 2**, Scheme 3). In this case, a chromatographic purification (SiO_2) proved to be necessary for the separation of allylic alcohol **5** and homoallylic alcohol **6**.



Scheme 3 Regiodivergent kinetic resolution of racemic 1,3-cycloheptadiene monoepoxide with Et_2Zn .

As regards symmetrical substrates, we recently reported that COT-monoepoxide (**3**), despite its non-planar tube form, can be successfully alkylated with appropriate organocopper reagents, without the occurrence of any ring-contraction-isomerization.^{6c} The enantioselective desymmetrization reaction of **3** with Et_2Zn can be improved with respect to our original report^{6c} and performed on a gram scale in the presence of catalytic amounts of $\text{Cu}(\text{OTf})_2$ / (R,R,R) -**1**¹² to give a clean crude reaction mixture containing (1*R*,4*R*)-4-ethyl-2,5,7-cyclooctatrienol (**7**) (**Procedure 3**, Scheme 4). Bis-allylic alcohol **7** can be obtained in a pure state by flash chromatography with a satisfactory 68% isolated yield and a high enantioselectivity of 94%.



Scheme 4 Catalytic enantioselective desymmetrization of COT-monoepoxide.

In summary, several optically active allylic and homoallylic alcohols can be synthesized by these procedures, allowing a flexible and practical gram-scale synthesis of this interesting class of compounds by the use of minimal amounts of BINOL-derived phosphoramidites.

All reactions were conducted in flame-dried glassware with magnetic stirring under an Ar atmosphere. Toluene was distilled from sodium/benzophenone ketyl and stored under Ar. ^1H NMR spectra were recorded on a Bruker AC-200 spectrometer. Chemical shifts are reported in ppm downfield from TMS with the solvent resonance as the internal standard (CDCl_3 : $\delta = 7.26$). ^{13}C NMR spectra were recorded on a Bruker AC-200 (50 MHz) spectrometer with complete proton decoupling. Chemical shifts are reported in ppm downfield from TMS with the solvent resonance as the internal standard (CDCl_3 : $\delta = 77.7$). Silica gel 60 (Macherey-Nagel 230–400 mesh) was used for flash chromatography.

(1*R*,4*R*)-(+)-4-Methyl-2-cyclohexen-1-ol (4): Kinetic Resolution Protocol (Procedure 1); Typical Procedure

A solution of $\text{Cu}(\text{OTf})_2$ (54.3 mg, 0.15 mmol) and (*R,R,R*)-**4** (0.160 g, 0.3 mmol) in anhyd toluene (30 mL) was stirred for 40 min. The colorless solution was cooled to -78°C and freshly distilled racemic 1,3-cyclohexadiene monoepoxide (**2b**) (4.80 g, 50 mmol) in anhyd toluene (5 mL) and isopropylbenzene (1.80 g as the internal standard) were added. Then a 2.0 M solution of Me_2Zn in toluene (11.4 mL) was added dropwise to the cooled (-78°C) reaction mixture over a period of 7 min. The reaction was monitored by GC and quenched with H_2O (15 mL) after 30 min (46% conversion). A light pale yellow suspension was obtained and vigorously stirred for 1 h at r.t. The precipitate was filtered through a glass sintered Büchner washing with the minimal amount of CH_2Cl_2 (3×8 mL). The organic layer was separated and the dried (MgSO_4) organic phase was distilled with a Vigreux column under reduced pressure (100 mm-Hg), thus obtaining a toluene solution (total volume 35 mL) containing the enantioenriched unreacted epoxide (1*S*,2*R*)-**2b**. The remaining liquid was distilled at 1.5 mmHg ($43\text{--}45^\circ\text{C}$) to give (1*R*,4*R*)-**4** (2.296 g, 41%), as a colorless liquid (contaminated with 4% regioisomeric homoallylic alcohol); $[\alpha]_{\text{D}^{20}} = +146$ ($c = 2.6$, CHCl_3). ^1H NMR, ^{13}C NMR, analytical data were previously described by us.^{4b}

Regiodivergent Kinetic Resolution Protocol (Procedure 2); Typical Procedure

A solution of $\text{Cu}(\text{OTf})_2$ (16.4 mg, 0.045 mmol) and (*R,R,R*)-**1** (49.1 mg, 0.091 mmol) in anhyd toluene (7.0 mL) was stirred at r.t. for 40 min. The solution was cooled to 0°C , and subsequently a solution of racemic allylic epoxide **2c** (1.0 g, 9.09 mmol) in toluene (1.0 mL) and 13.6 mmol of Et_2Zn (12.4 mL of a 1.1 M solution in toluene) were added. The reaction mixture was maintained at 0°C and quenched with 5% aq HCl (10.0 mL) after 1 h (>98% conversion). Extraction with Et_2O (3×30 mL) and evaporation of the dried (MgSO_4) organic phase gave a crude product containing allylic alcohol **5** (53%) and homoallylic alcohol **6** (47%), which was subjected to flash chromatography on SiO_2 (3×19 cm column) eluting with hexanes (800 mL) containing 20% EtOAc, to give, as the second eluting fractions, pure (1*R*,4*R*)-4-ethyl-2-cyclohepten-1-ol (**5**) (534 mg, 42%), as a colorless liquid. The enantiomeric excess of **5** (83%) was determined by chiral GC (β -cyclodextrin column, isothermal 110°C); t_{R} 43.82 min (minor), t_{R} 44.36 min (major); $[\alpha]_{\text{D}^{20}} = +101$ ($c = 1.5$, MeOH); $R_{\text{f}} = 0.20$ (20% EtOAc in hexanes).

^1H NMR: $\delta = 5.43\text{--}5.77$ (m, 2 H), 4.29–4.41 (m, 1 H, *CHOH*), 2.05–2.23 (m, 1 H), 1.21–1.89 (m, 8 H), 0.90 (t, $J = 7.2$ Hz, 3 H).

^{13}C NMR: $\delta = 136.5$, 135.3, 70.9, 40.2, 36.3, 32.2, 29.1, 22.7, 12.3.

MS (EI, 70 eV): m/z (%) = 122 (91), 107 (17), 93 (38), 81 (100), 67 (39), 39 (48).

Anal. Calcd for $\text{C}_9\text{H}_{16}\text{O}$: C, 77.09; H, 11.50. Found: C, 76.88; H, 11.39.

The first eluting fractions of the above-described flash chromatography afforded pure (1*S*,2*S*)-2-ethyl-3-cyclohepten-1-ol (**6**) (483 mg, 38%), as a colorless liquid. The enantiomeric excess of **6** (>95%) was determined by chiral GC (β -cyclodextrin column, isothermal 110°C); t_{R} 34.06 min (minor), t_{R} 34.56 min (major); $[\alpha]_{\text{D}^{20}} = +58$ ($c = 2.0$, MeOH); $R_{\text{f}} = 0.26$ (20% EtOAc in hexanes).

^1H NMR: $\delta = 5.81\text{--}5.97$ (m, 1 H), 5.40–5.52 (m, 1 H), 3.54 (dt, $J = 8.0$, 3.1 Hz, 1 H), 2.26–2.36 (m, 1 H), 1.95–2.15 (m, 2 H), 1.27–1.82 (m, 6 H), 0.93 (t, $J = 7.4$ Hz, 3 H).

^{13}C NMR: $\delta = 133.7$, 132.7, 71.7, 47.8, 39.0, 29.0, 25.0, 24.0, 11.7.

MS (EI, 70 eV): m/z (%) = 122 (62), 111 (43), 93 (100), 79 (89), 67 (47), 55 (57).

Anal. Calcd for $\text{C}_9\text{H}_{16}\text{O}$: C, 77.09; H, 11.50. Found: C, 77.21; H, 11.33.

Desymmetrization of COT Monoepoxide (Procedure 3):

(1*R*,4*R*)-4-Ethyl-2,5,7-cyclooctatrien-1-ol (7); Typical Procedure

A solution of $\text{Cu}(\text{OTf})_2$ (54.0 mg, 0.15 mmol) and chiral ligand (*R,R,R*)-**1** (162.0 mg, 0.3 mmol) in anhyd toluene (5 mL) was stirred at r.t. for 40 min. The solution was cooled to -78°C and a solution of freshly distilled **3^{9b}** (containing 7% of unreacted COT, 1.30 g, ca. 10.0 mmol) in toluene (1 mL) was added. After 5 min., Et_2Zn (13.6 mL, 15 mmol) was added and the stirred reaction mixture was allowed to warm slowly up to 0°C . The mixture was quenched after 3 h (>95% conversion) at 0°C with sat. aq NH_4Cl solution (10 mL). Extraction with Et_2O (3×35 mL) and evaporation of the dried (MgSO_4) organic phase afforded a crude liquid reaction mixture (1.40 g) essentially made up of compound **7** and chiral ligand **1**. Compound **7** can be obtained in a pure state (1.020 g, 68% yield) as a pale yellow oil by flash chromatography (SiO_2) eluting with hexanes containing 15% of EtOAc and 2% Et_3N . The enantiomeric excess (94%) can be determined on the corresponding saturated *trans*-4-ethyl-cyclooctanol ($\text{H}_2/\text{PtO}_2/\text{MeOH}$) as reported by us before^{6c}; $[\alpha]_{\text{D}^{20}} = +325$ ($c = 1.0$, MeOH); $R_{\text{f}} = 0.27$ (20% EtOAc in hexanes).

^1H NMR: $\delta = 6.07\text{--}6.17$ (m, 2 H), 5.55–5.61 (m, 1 H), 5.15–5.43 (m, 3 H), 4.82–4.93 (m, 1 H, *CHOH*), 2.65–2.75 (m, 1 H), 1.51–1.65 (m, 2 H), 0.90 (t, $J = 7.3$ Hz, 3 H).

^{13}C NMR: $\delta = 133.5$, 133.0, 132.6, 131.5, 128.3, 127.0, 70.5, 39.6, 29.6, 12.4.

Anal. Calcd for $\text{C}_{10}\text{H}_{14}\text{O}$: C, 79.96; H, 9.39. Found: C, 78.96; H, 8.76.

Acknowledgment

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